

BSMA GeoAI — Borewell & Groundwater Siting Report

Applied GeoAI Division | Hard-Rock Groundwater Prospecting (Musi Sub-Basin, Telangana)

Report ID: BSMA-GW-2026-KF02
Date: 2026-08-25

Status: Validated & Sited

1. Land Parcel & Hydrogeological Profile

Land Parcel:	Mango Farm (Karun Farmland 1)	Total Area:	20.37 Acres (8.242 ha)
Centroid Coordinates:	17.43488°N, 79.08173°E	Potential Index:	65.9 / 100 (High Potential)
Lithology / Terrain:	Weathered Granite-Gneiss Saprolite	Rainfall Recharge:	~820 mm/year (SW Monsoon)

2. Farm Siting Plan & Candidate Locations Map

Hydrogeological Siting Summary

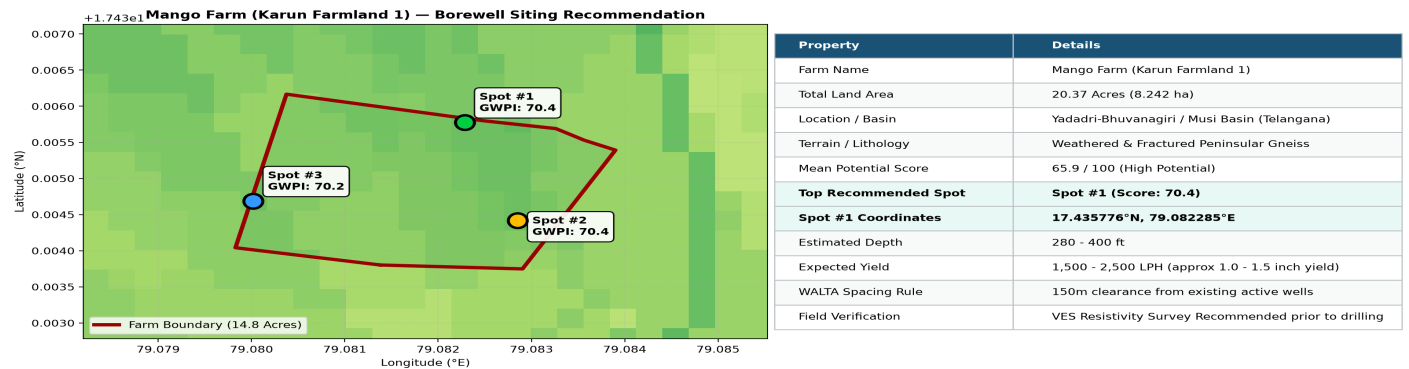


Figure 1: High-resolution plot boundary with color-coded candidate drilling spots (#1 Primary, #2 Secondary, #3 Alternative) and GWPI heatmap overlay.

3. Ranked Candidate Drilling Spots & Technical Specifications

Rank & Priority	Coordinates (Lat, Lon)	GWPI Score	Est. Depth	Expected Yield	Elevation / Slope
Spot #1 Primary	17.43578°N 79.08228°E	70.4/100 Moderate to High Potential	280 - 400 ft	1,500 - 2,500 LPH (approx 1.0 - 1.5 inch yield)	336.3m MSL (2.5% slope)
Spot #2 Secondary	17.43442°N 79.08285°E	70.4/100 Moderate to High Potential	280 - 400 ft	1,500 - 2,500 LPH (approx 1.0 - 1.5 inch yield)	332.3m MSL (2.4% slope)
Spot #3 Alternative	17.43469°N 79.08002°E	70.2/100 Moderate to High Potential	280 - 400 ft	1,500 - 2,500 LPH (approx 1.0 - 1.5 inch yield)	331.8m MSL (4.3% slope)

4. Geological Rationales & Drilling Recommendations

- Spot #1 (17.43578°N, 79.08228°E):** Located on a gentle slope (2.5%) with favorable fracture density and high moisture convergence in the weathered granite zone. Recommended drilling: 6.5" DTH hammer with 40–60 ft casing through weathered saprolite.
- Spot #2 (17.43442°N, 79.08285°E):** Located on a gentle slope (2.4%) with favorable fracture density and high moisture convergence in the weathered granite zone. Recommended drilling: 6.5" DTH hammer with 40–60 ft casing through weathered saprolite.
- Spot #3 (17.43469°N, 79.08002°E):** Located on a gentle slope (4.3%) with favorable fracture density and high moisture convergence in the weathered granite zone. Recommended drilling: 6.5" DTH hammer with 40–60 ft casing through weathered saprolite.

5. Regional Catchment & Drainage Fracture Context

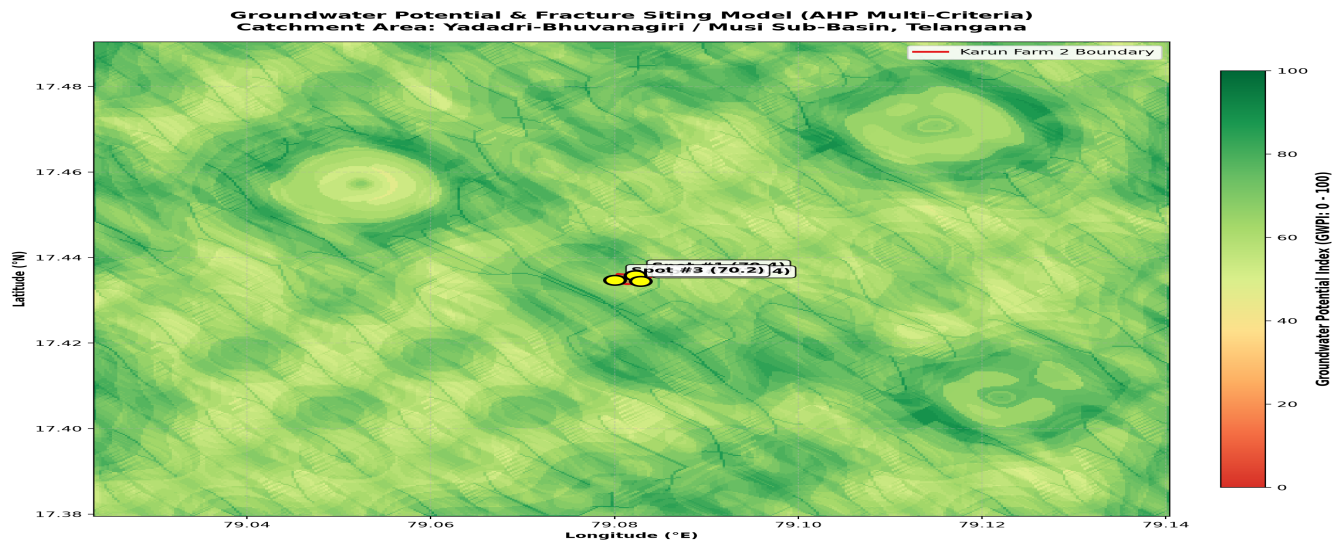


Figure 2: Regional 6km catchment hydrogeological model showing secondary NW-SE / NE-SW lineament corridors and Musi sub-basin drainage network.

6. AHP Multi-Criteria Thematic Weights (Hard-Rock Terrain)

Geology / Lithology: 25.72%	Fracture Lineaments: 20.87%	Topographic Slope: 15.75%	Drainage Density: 12.01%
Wetness Index (TWI): 9.12%	Land Use / LULC: 6.96%	Soil Infiltration: 5.38%	Rainfall Recharge: 4.19%

7. Landowner Siting Guidance & Field Action Plan

Hydrogeological Siting Summary:	Mango Farm (Karun Farmland 1) shows an overall High Potential (Average GWPI: 65.9/100). Secondary fracture corridors and moisture convergence zones are concentrated along the central drainage axis. Spot #1 (Score: 70.4/100) is the top-ranked drilling target.
Pre-Drilling Action Steps:	1. Conduct 1D/2D VES geophysical survey at Spot #1 to verify fracture depth (280–400 ft). 2. Confirm 150m boundary clearance from neighboring active borewells under WALTA guidelines. 3. Provide this technical sheet to the drilling contractor before rig mobilization.

8. Statutory Clearances & Drilling Rig Operator Protocol

✓ Telangana WALTA Act Compliance:	All candidate spots maintain >= 150m spacing. Confirm 150m clearance from neighbor wells.
■ Mandatory VES Resistivity Survey:	Vertical Electrical Sounding (VES) or 2D Resistivity Imaging required at Spot #1 prior to drilling rig mobilization.
■■ Drilling Specifications:	Minimum 40–60 ft MS/PVC casing through weathered saprolite until hard bedrock is seated; 6.5" DTH hammer drill bit with 900–1100 CFM / 300 PSI compressor.